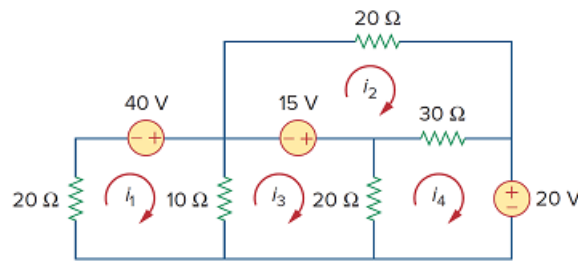


## Problem

By inspection, write the mesh-current equations for the circuit in Fig. 3.116.

**Figure 3.116:**



## Step-by-step solution

## Step 1 of 12

Refer to the circuit in Figure 3.116 in the textbook.

If a circuit has  $N$  meshes, the mesh-current equations can be expressed in terms of resistances:

$$\begin{pmatrix} R_{11} & \cdots & R_{1N} \\ \vdots & \ddots & \vdots \\ R_{N1} & \cdots & R_{NN} \end{pmatrix} \begin{pmatrix} i_1 \\ \vdots \\ i_N \end{pmatrix} = \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix} \text{ or simply } \mathbf{Ri} = \mathbf{v}.$$

Where,

Sum of the resistances in mesh  $k$  is  $R_{kk}$ ,

Negative of the sum of the resistances in common with meshes  $k$  and  $j$ ,  $k \neq j$  is,

$$R_{kj} = R_{jk}$$

Unknown mesh current for mesh  $k$  in the clockwise direction is  $i_k$ .

Sum taken clockwise of all independent voltage sources in mesh  $k$ , with voltage rise treated as positive is  $v_k$ .

## Step 2 of 12

The circuit in Figure 1 has five meshes, so we need five mesh equations. This implies that the size of the resistance matrix  $\mathbf{R}$ , is 4 by 4.

Each of the diagonal terms is the sum of the resistances in the related mesh, while each of the off-diagonal terms is the negative of the resistances common to meshes.

Calculate the diagonal terms of  $\mathbf{R}$ , in ohms:

Calculation of diagonal terms:

Calculate the value of  $R_{11}$ .

$$\begin{aligned} R_{11} &= 20 + 10 \\ &= 30 \end{aligned}$$

Therefore, the value of the  $R_{11}$  is 30.

## Step 3 of 12

Calculate the value of  $R_{22}$ .

$$\begin{aligned} R_{22} &= 20 + 30 \\ &= 50 \end{aligned}$$

Therefore, the value of the  $R_{22}$  is 50.

**Step 4** of 12

Calculate the value of  $R_{33}$ .

$$\begin{aligned} R_{33} &= 20 + 10 \\ &= 30 \end{aligned}$$

Therefore, the value of the  $R_{33}$  is 30.

**Step 5** of 12

Calculate the value of  $R_{44}$ .

$$\begin{aligned} R_{44} &= 20 + 30 \\ &= 50 \end{aligned}$$

Therefore, the value of the  $R_{44}$  is 50.

**Step 6** of 12

Calculation of non-diagonal terms:

Calculate the value of  $R_{12}$ .

$$R_{12} = 0$$

Therefore, the value of the  $R_{12}$  is 0.

The value of the  $R_{21}$  is equal to 0 since  $R_{21}$  is equal to  $R_{12}$ .

**Step 7** of 12

Calculate the value of  $R_{13}$ .

$$R_{13} = -10$$

Therefore, the value of the  $R_{13}$  is -10.

Since  $R_{31}$  is equal to  $R_{13}$ ,

$$R_{31} = -10$$

Thus, the value of the  $R_{31}$  is -10.

**Step 8** of 12

Calculate the value of  $R_{23}$ .

$$R_{23} = 0$$

Therefore, the value of the  $R_{23}$  is 0.

Since  $R_{23}$  is equal to  $R_{32}$ ,

$$R_{32} = 0$$

Thus, the value of the  $R_{23}$  is 0.

**Step 9** of 12

Calculate the value of  $R_{24}$ .

$$R_{24} = -30$$

Therefore, the value of the  $R_{24}$  is -30.

Since  $R_{42}$  is equal to  $R_{24}$ ,

$$R_{42} = -30$$

Thus, the value of the  $R_{42}$  is -30.

**Step 10** of 12

Calculate the value of  $R_{34}$ .

$$R_{34} = -20$$

Therefore, the value of the  $R_{34}$  is -20.

Since  $R_{43}$  is equal to  $R_{34}$ ,

$$R_{43} = -20$$

The value of the  $R_{43}$  is -20.

**Step 11** of 12

Calculate the value of  $R_{14}$ .

$$R_{14} = 0$$

Therefore, the value of the  $R_{14}$  is 0.

Since  $R_{41}$  is equal to  $R_{14}$ ,

$$R_{41} = 0$$

The value of the  $R_{41}$  is 0.

**Step 12** of 12

Write the node voltage equations in matrix representation.

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ R_{21} & R_{22} & R_{23} & R_{24} \\ R_{31} & R_{32} & R_{33} & R_{34} \\ R_{41} & R_{42} & R_{43} & R_{44} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

Substitute  $-10$  for  $R_{13}$  and  $R_{31}$ ,  $0$  for  $R_{23}$  and  $R_{32}$ ,  $0$  for  $R_{41}$  and  $R_{14}$ ,  $-20$  for  $R_{43}$  and  $R_{34}$ ,  $0$  for  $R_{21}$  and  $R_{12}$ ,  $-30$  for  $R_{42}$  and  $R_{24}$ ,  $30$  for  $R_{11}$ ,  $50$  for  $R_{22}$ ,  $30$  for  $R_{33}$ ,  $50$  for  $R_{44}$ ,  $40$  for  $v_1$ ,  $-15$  for  $v_2$ ,  $15$  for  $v_3$ , and  $-20$  for  $v_4$ .

$$\begin{bmatrix} 30 & 0 & -10 & 0 \\ 0 & 50 & 0 & -30 \\ -10 & 0 & 30 & -20 \\ 0 & -30 & -20 & 50 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 40 \\ -15 \\ 15 \\ -20 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & -1 & 0 \\ 0 & 5 & 0 & -3 \\ -1 & 0 & 3 & -2 \\ 0 & -3 & -2 & 5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 4 \\ -1.5 \\ 1.5 \\ -2 \end{bmatrix}$$

Therefore, the set of mesh current equations for the given circuit is,

$$\begin{bmatrix} 3 & 0 & -1 & 0 \\ 0 & 5 & 0 & -3 \\ -1 & 0 & 3 & -2 \\ 0 & -3 & -2 & 5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 4 \\ -1.5 \\ 1.5 \\ -2 \end{bmatrix}$$